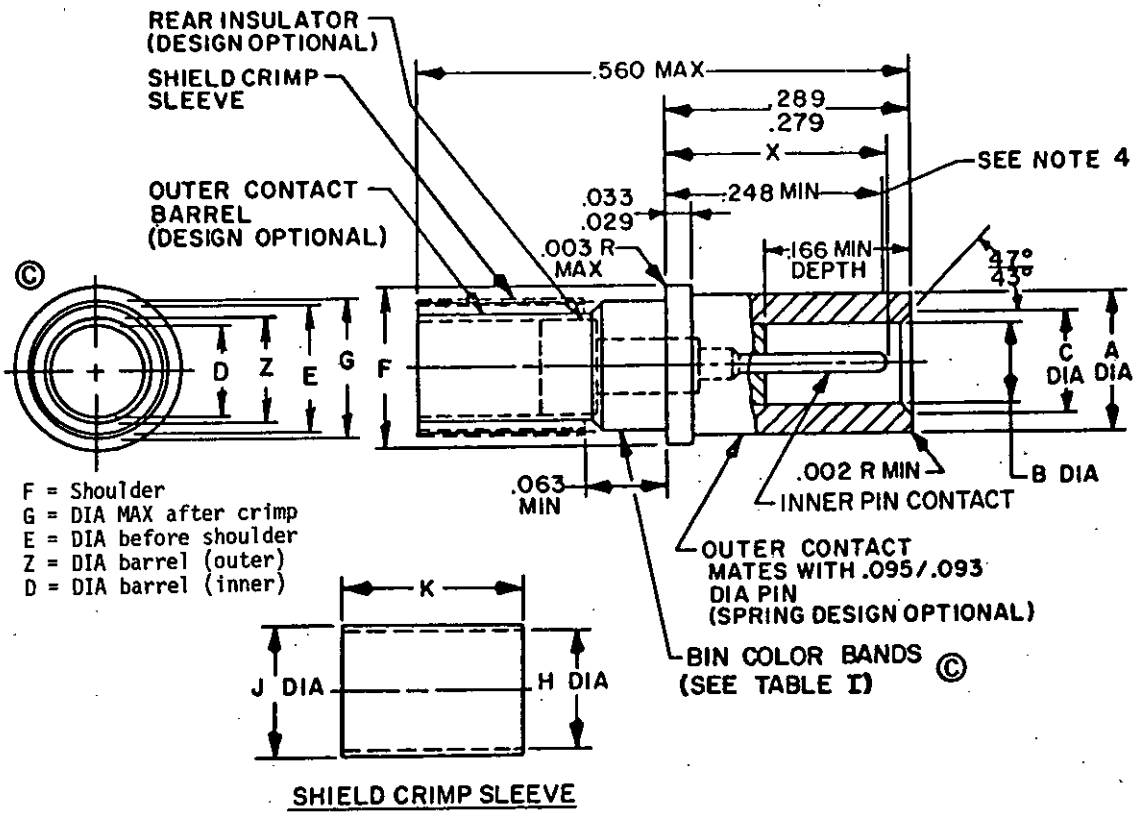
MILITARY SPECIFICATION SHEET

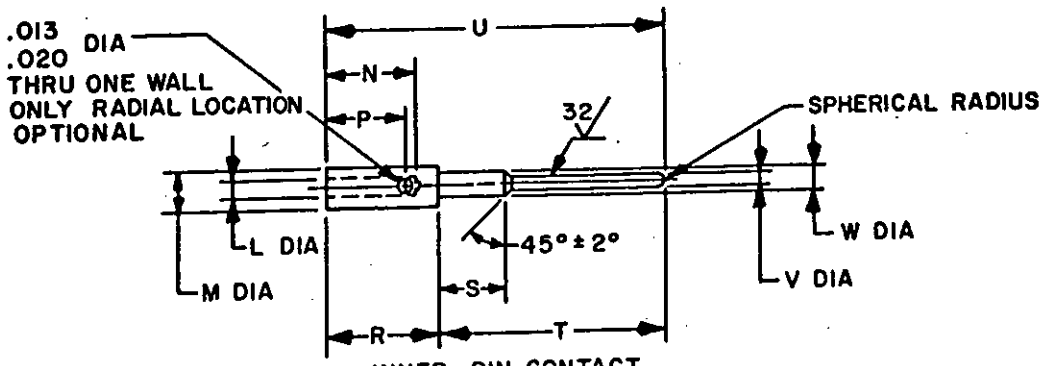
CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-C-38999 SERIES II CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.



SHIELD CRIMP SLEEVE



INNER PIN CONTACT

© Denotes Changes

FIGURE 1. Socket contacts.

FSC 5935

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BIN code	A Dia.	B Dia.	C Dia.	D Dia. Min.	E Dia.	F Dia.	G Dia. Max.	H Dia. Min.	J Dia. Max.	K	L Dia. Min.	M Dia. Max.
210	.161	.100 .097	.123 .118	.090	.151 .148	.182 .179	.156	.127	.169	.125 .115	.0225	.052
402	↓	↓	↓	.108	↓	↓	↓	.144	↓	↓	.0225	↓
403	↓	↓	↓	.108	↓	↓	↓	.144	↓	↓	.0355	↓
404	↓	↓	↓	.090	↓	↓	↓	.127	↓	↓	.0355	↓
405	↓	↓	↓	.090	↓	↓	↓	.127	.169	↓	.0225	↓
406	↓	↓	↓	.117	↓	↓	↓	.156	.174	↓	.0270	↓
407	↓	↓	↓	.090	↓	↓	↓	.144	.169	↓	.0225	↓
408	.161	.100 .097	.123 .118	.108	.151 .148	.182 .179	.156	.156	.174	.125 .115	.0355	.052

BIN code	N Min.	P	R	S	T	U REF.	V Dia.	W Dia.	X	Z Dia. Max.
210	.112	.103 .096	.146 .140	.039 .033	.222 .219	.3635	.0205 .0195	.035 .033	.276 .266	.110
402	↓	↓	↓	↓	↓	↓	↓	↓	↓	.127
403	↓	↓	↓	↓	↓	↓	↓	↓	↓	.127
404	↓	↓	↓	↓	↓	↓	↓	↓	↓	.110
405	↓	↓	↓	↓	↓	↓	↓	↓	↓	.110
406	↓	↓	↓	↓	↓	↓	↓	↓	↓	.136
407	↓	↓	↓	↓	↓	↓	↓	↓	↓	.110
408	.112	.103 .096	.146 .140	.039 .033	.222 .219	.3635	.0205 .0195	.035 .033	.276 .266	.127

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.033	.84	.096	2.44	.118	3.00	.151	3.84	.219	5.56
.003	.08	.035	.89	.097	2.46	.123	3.12	.156	3.96	.222	5.64
.013	.33	.0355	.902	.100	2.54	.125	3.18	.158	4.01	.248	6.30
.0195	.495	.039	.99	.103	2.62	.127	3.23	.161	4.09	.266	6.76
.020	.51	.052	1.32	.108	2.74	.136	3.45	.166	4.22	.276	7.01
.0205	.521	.063	1.60	.110	2.79	.140	3.56	.169	4.29	.279	7.09
.0225	.572	.090	2.29	.112	2.84	.144	3.66	.174	4.42	.289	7.34
.027	.69	.093	2.36	.115	2.92	.146	3.71	.179	4.55	.3635	9.233
.029	.74	.095	2.41	.117	2.97	.148	3.76	.182	4.62	.560	14.22

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the outer contact spring. Provision for clearance hole shall be provided.
5. Crimp deformation: The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed G diameter.

FIGURE 1. Socket contacts - Continued.

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REQUIREMENTS:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Mating contacts: MIL-C-39029/28.

Tools: See table II.

TABLE I. Design characteristics.

BIN code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
210	Red	Brown	Black	1/2/ M17/119-RG174 M17/113-RG316 M17/094-RG179 Times AA3248 Teledyne 11299 Thermax 75-738-BCCWXE Tensolite 30888/L707YX-1 Haveg 8100207	12	D	B
402	Yellow	Black	Red	1/ M17/095-RG180 Raychem 9527D1514-2L Raychem 9528A1318 Microdot 293-3922			
403	Yellow	Black	Orange	1/ Microdot 250-4070			
404	Yellow	Black	Yellow	1/ Raychem 48-502 & 5022E5111			
405	Yellow	Black	Green	1/ Raychem 48-950 & 9530D5117			
406	Yellow	Black	Blue	1/ Raychem 7624D1311 Raychem 9527A1318			
407	Yellow	Black	Violet	1/ Gore GWN1159A			
408	Yellow	Black	Gray	1S50MJ-16, -20, -40, -70 (MIL-C-915/42)	↓	↓	↓

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

© TABLE II. Tools.

BIN code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Positioner		
210, 402, 403, 404, 405, 406, 407, 408	M22520/2-01	M22520/2-34	M22520/31-01	M22520/31-02	M81969/8-09 or M81969/14-04	M81969/8-10 or M81969/14-04

© Cable to contact information: See table III.

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© TABLE III. Cable to contact information.

BIN code	Cable accommodated	Inner contact tool selector setting no.
210	M17/119-RG174	4
	M17/113-RG316	5
	M17/094-RG179	3
	1/ 2/ Times AA3248	4
	Teledyne 11299	4
	Thermax 75-738-BCCWXE	4
	Tensolite 30888/L707YX-1	4
	Haveg 8100207	4
402	M17/095-RG180	3
	1/ Raychem 9527D1514-2L	5
	Raychem 9528A1318	4
	Microdot 293-3922	3
403	1/ Microdot 250-4070	4
404	1/ Raychem 48-502 & 5022E5111	4
405	1/ Raychem 48-950 & 9530D5117	3
406	1/ Raychem 7624D1311	5
	Raychem 9527A1318	5
407	1/ Gore GWN1159A	4
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	4

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Contact resistance: See table IV.
Test current:
Inner contact - 1 ampere.
Outer contact - 12 amperes.

Low signal level contact resistance (inner contact only): See table V.

Contact engagement and separation forces (socket contact only): The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in the following, and surface roughness shall not exceed 3 microinches. Provision for clearance hole shall be provided.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0950 $^{+.0002}_{-.0000}$	NA	NA	30	36	NA
.0930 $^{+.0000}_{-.0002}$	3.0	2.5	NA	NA	NA

Dielectric withstanding voltage (applied between inner and outer contact):
Test voltage:
At sea level - 1000 Vac rms.
At 50,000 feet - 250 Vac rms.

Tensile strength (inner and outer contact crimp joint): See table V.

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TABLE IV. Contact resistance.

BIN code	Cable accommodated	Maximum voltage drop (millivolts) 25° +3° -0°C Inner contact	25° +3° -0°C Outer contact	2/ 25° +3° -0°C Inner contact	2/ 25° +3° -0°C Outer contact	200° +3° -0°C Inner contact	200° +3° -0°C Outer contact	Maximum average voltage drop
210	1/3/ M17/119-RG174	55	85	66	102	94 4/	145 4/	NA
	M17/113-RG316	55	75	66	90	94	128	
	M17/094-RG179	120	70	144	84	204	119	
	Times AA3248	170	150	204	180	290	255	
	Teledyne 11299	↓	↓	↓	↓	↓	↓	
	Thermax 75-738-BCCWXE							
	Tensolite 30888/L707YX-1							
	Haveg 8100207							
402	1/ M17/095-RG180	120	60	144	72	204	102	
	Raychem 9527D1514-2L					204 5/	102 5/	
	Raychem 9528A1318					204 5/	102 5/	
	Microdot 293-3922					204	102	
403	1/ Microdot 250-4070	↓		↓		204	102	
404	1/ Raychem 48-502 & 5022E5111	55		66		94 5/	102 5/	
405	1/ Raychem 48-950 & 9530D5117	120		144		204 5/	102 5/	
406	1/ Raychem 7624D1311	55		66		94 5/	102 5/	
	Raychem 9527A1318	120		144		204 5/	102 5/	
407	1/ Gore GWN1159A	120		144		204	102	
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	↓	66	↓	94 4/	102 4/	↓

1/ Or equivalent.
2/ After conditioning.
3/ High tensile strength copper alloy wire.
4/ 85° +3° -0°C.
5/ 115° +3° -0°C.

© TABLE V. Low signal level contact resistance (inner contact only) and tensile strength.

BIN code	Cable accommodated	Maximum contact resistance (milliohms) Initial	After conditioning	Tensile load (pounds) (minimum) Inner contact	Outer contact
210	1/2/ M17/119-RG174	55	66	15.0	15.0
	M17/113-RG316	55	66	10.0	15.0
	M17/094-RG179	120	144	3.5	15.0
	Times AA3248	170	204	7.0	35.0
	Teledyne 11299	↓	↓	↓	↓
	Thermax 75-738-BCCWXE				
	Tensolite 30888/L707YX-1				
	Haveg 8100207				
402	1/ M17/095-RG180	120	144	3.5	20.0
	Raychem 9527D1514-2L			6.0	
	Raychem 9528A1318			6.0	
	Microdot 293-3922			3.5	
403	1/ Microdot 250-4070	↓	↓	4.0	
404	1/ Raychem 48-502 & 5022E5111	55	66	10.0	
405	1/ Raychem 48-950 & 9530D5117	120	144	3.5	
406	1/ Raychem 7624D1311	55	66	6.0	
	Raychem 9527A1318	120	144	6.0	
407	1/ Gore GWN1159A	120	144	4.0	
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	66	15.0	↓

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

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© Vibration: Test condition VI, letter J, per method 2005 of MIL-STD-1344. Except directions shall be longitudinal and perpendicular. Time duration shall be 8 hours in each direction.

© Shock (specified pulse): Test condition D, per method 2004 of MIL-STD-1344.

© Shipping: Figure 1 illustrates final assembly; parts are to be shipped unassembled in a suitable container. Manufacturers recommended assembly instructions shall be shipped with the unit package.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Military part number: See table VI.

© TABLE VI. Military part number.

BIN code	Military part number
210	M39029/27-12A
402	M39029/27-12B
403	M39029/27-12C
404	M39029/27-12D
405	M39029/27-12E
406	M39029/27-12F
407	M39029/27-12G
408	M39029/27-12H

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

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